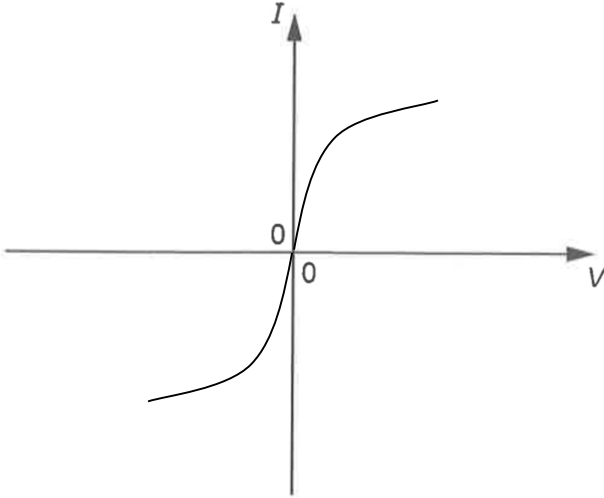
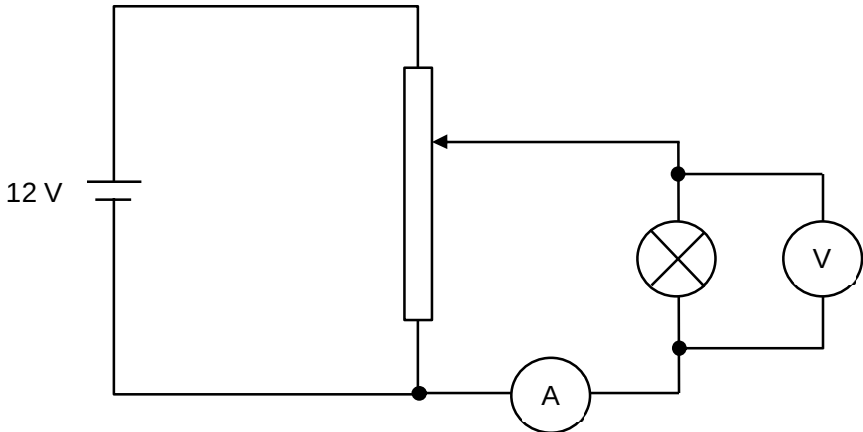


	single wire.
3(a)(i)	Resistance = $\frac{2.90}{0.968} = 3.00 \, \Omega$
3(a)(ii)	Two resistors: $E = 2.90 + 2.90 + 0.968r$ Three resistors: $E = 1.277 \left(3.00 + \frac{3.00}{2} + r \right)$ Therefore $2.90 + 2.90 + 0.968r = 1.277 \left(3.00 + \frac{3.00}{2} + r \right)$ $r = 0.176 \, \Omega$
3(a)(iii)	$E = 2.90 + 2.90 + 0.968(0.176) = 5.97 \, \text{V}$
3(b)(i)	 Fig. 3.2

No.	Solution
3(b)(ii)	The potential difference across the filament increases. The current through the filament increases. The temperature of the filament rises, leading to a greater rate of collision between metal ions and electrons. This causes the resistance of the filament to rise.
3(b)(iii)	The resistance of the lamp cannot be deduced from $\frac{1}{\text{gradient of } I-V \text{ characteristic}}$. Resistance is calculated from the ratio of the potential difference across the lamp to the current flowing through the lamp.
3(c)	
4(a)(i)	Gravitational potential energy converts to kinetic energy